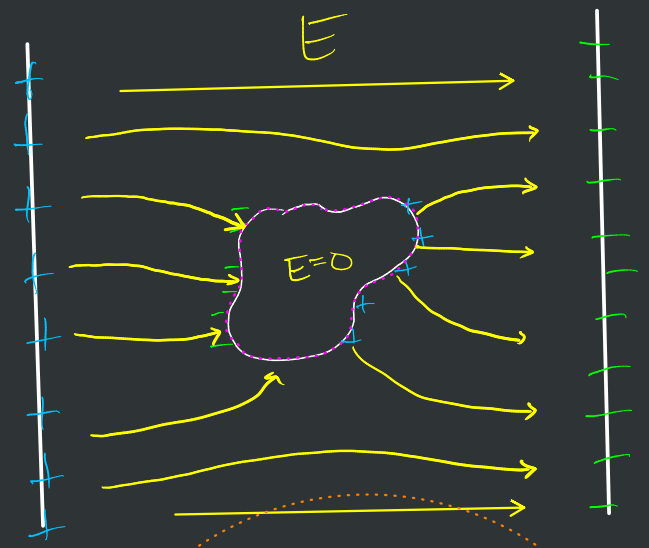
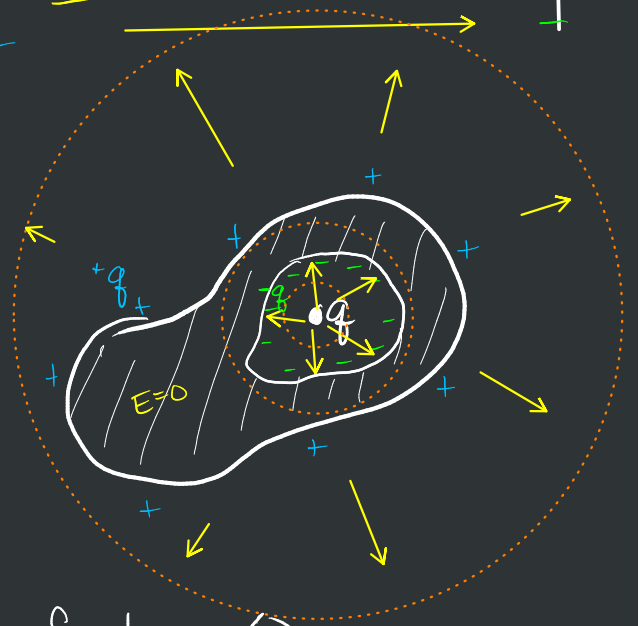
Chapter 3

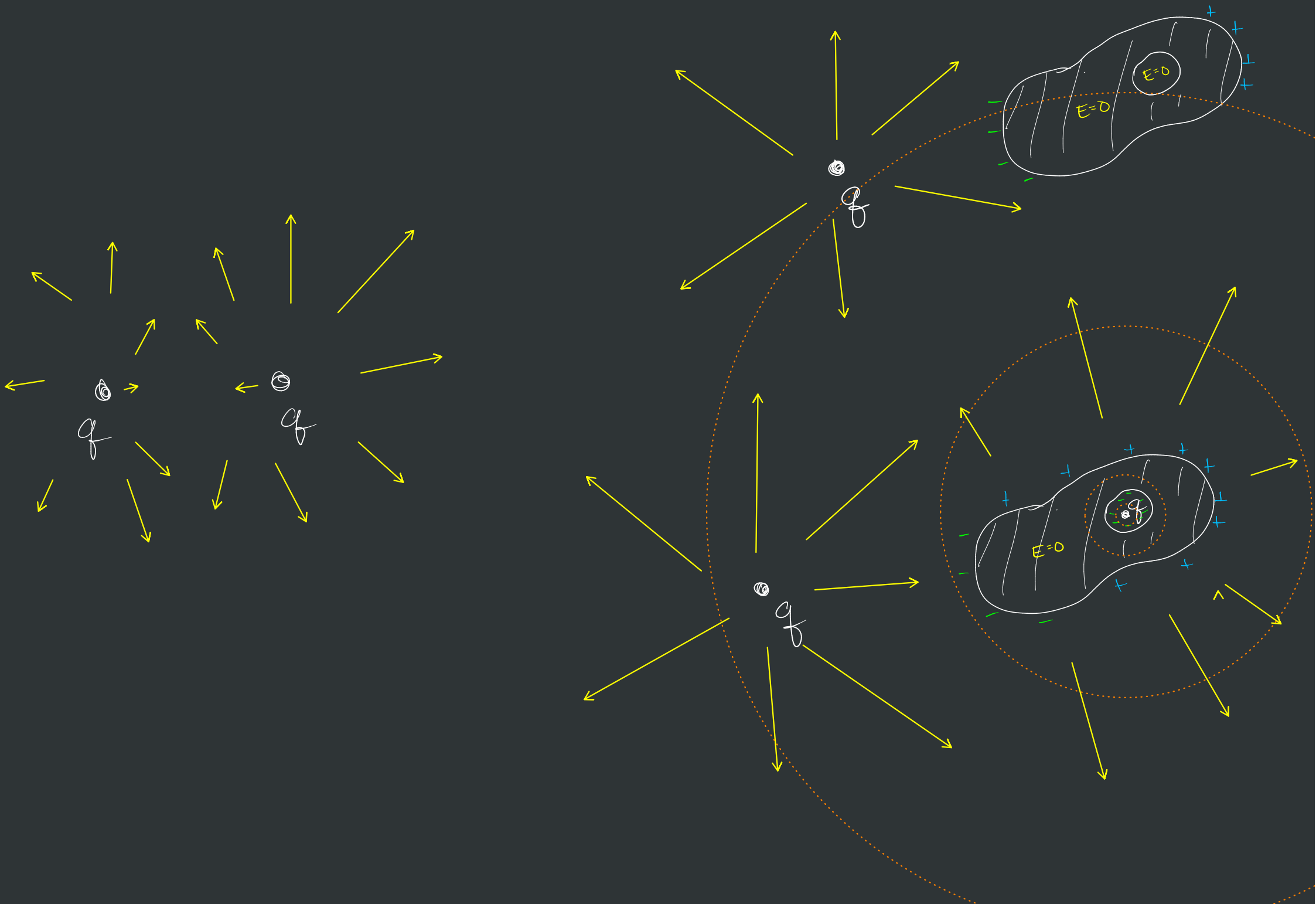
Conductors vs. **insulators** → shift of charges but mass movement of charge.

→ charges move freely (no work) through the bulk of the material



- in the presence of an electric field charges move w/i the conductor until the field inside is zero.
- so the charge reside on the surface
- so the surface is an equipotential
- electric field just outside the conductor is perpendicular to the surface
- at the surface Gauss's Law is satisfied for the total electric field $\epsilon_0 \oint_S \vec{E} \cdot d\vec{a} = \int \sigma \, da = Q_{NET}$




q
q
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E=0
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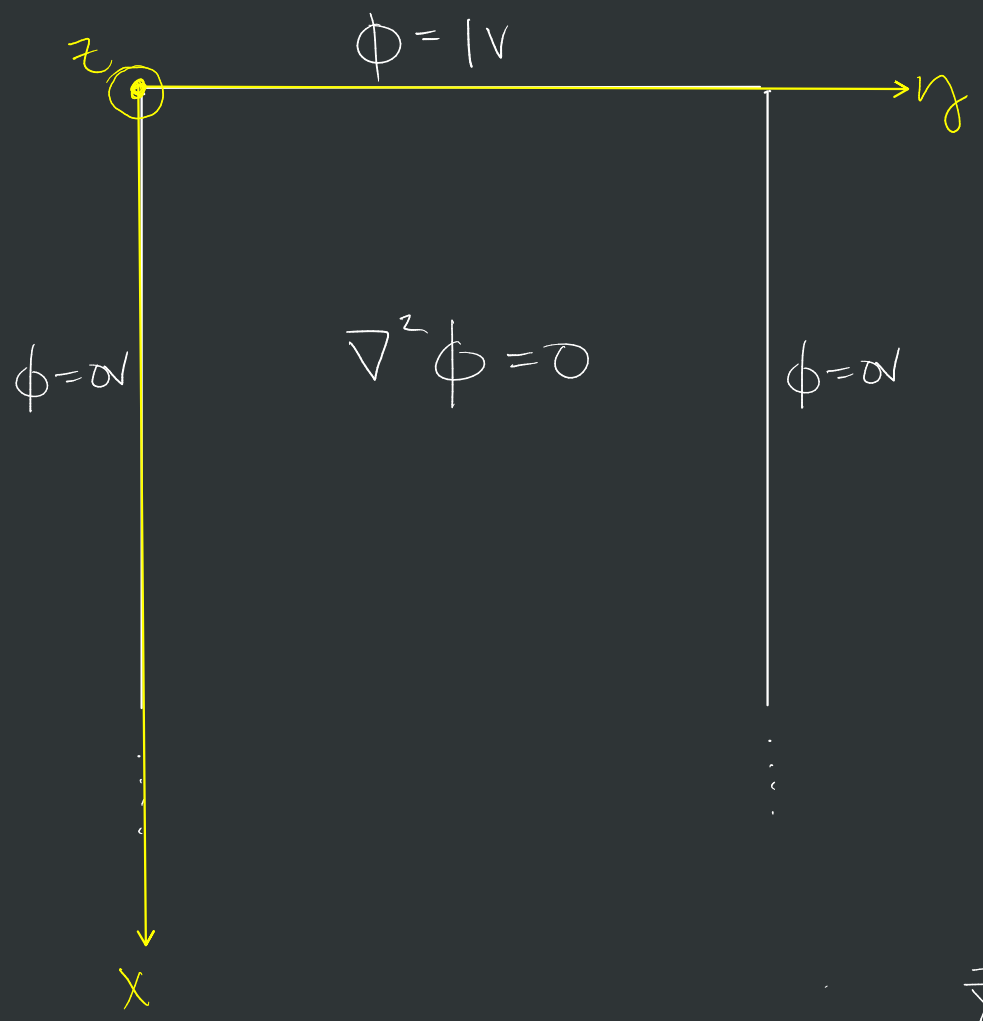

in cartesian

$$\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} + \underbrace{\frac{\partial^2 \phi}{\partial z^2}}_{=0} = 0$$

$$\phi = X(x) \cdot Y(y)$$

$$\frac{\partial^2 X(x) Y(y)}{\partial x^2} + \frac{\partial^2 X Y}{\partial y^2} = 0$$

$$\frac{1}{X \cdot Y} \left[Y \frac{\partial^2 X}{\partial x^2} + X \frac{\partial^2 Y}{\partial y^2} = 0 \right] \frac{1}{X \cdot Y}$$

$$\underbrace{\frac{1}{X} \frac{\partial^2 X}{\partial x^2}}_{= k^2} + \underbrace{\frac{1}{Y} \frac{\partial^2 Y}{\partial y^2}}_{= -k^2} = 0$$

$$\frac{1}{X}\frac{\partial^2 X}{\partial x^2} = -k^2$$

$$\frac{\partial^2 X}{\partial x^2} = -k^2 X$$

$$\frac{1}{Y}\cdot\frac{\partial^2 Y}{\partial y^2} = k^2$$

$$\frac{\partial^2 Y}{\partial y^2} = k^2 Y$$